QUESTION 1

Given $f(x) = 5 - 3x$.

(a) Evaluate the following:

(1) $f(-1)$ (1)

$= 5 - 3(-1)$
$= 8$ ✓

(2) $3 \cdot f(k)$ (2)

$3.(5 - 3k)$
$= 15 - 9k$
✓ ✓

(3) $f(f(0))$ (2)

$f(0) = 5 - 3(0)$
$= 5$ ✓
$f(5) = 5 - 3(5)$
$= 5 - 15$
$= -10$ ✓ ✓

(b) Describe in words how f was transformed to become $y = -3 - 3x$. (2)

$y = -3x - 3$

Shifted ✓
Translated/ down 8 ✓
units

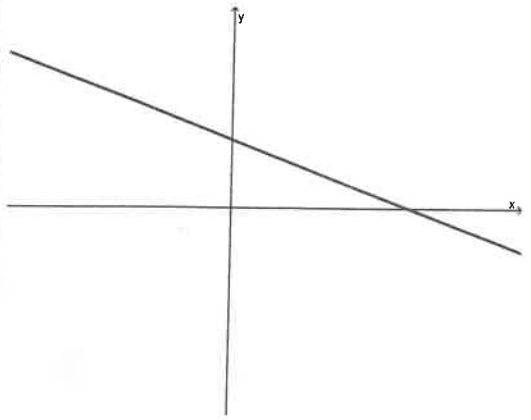
[7]

QUESTION 2

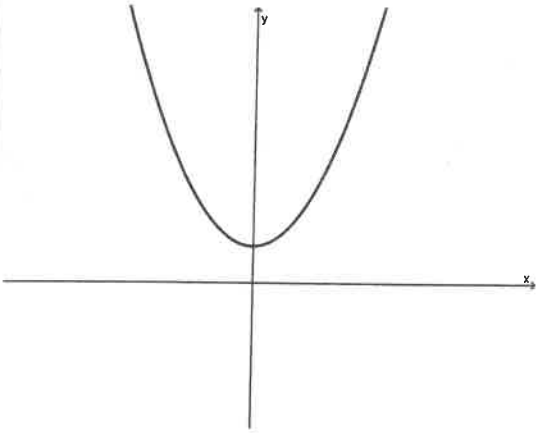
(a) Match each graph to the correct description of its equation.
Options may be used once only.

(3)

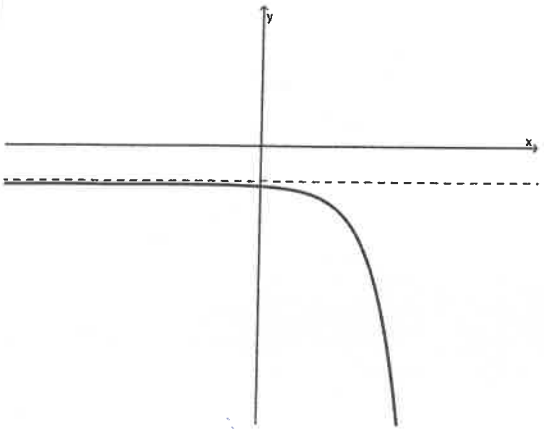
Graph 1: $y = ax + q$



Graph 2: $y = ax^2 + q$



Graph 3: $y = a \cdot b^x + q$



Options	
A:	$a>0$ and $q>0$
B:	$a<0$ and $q<0$
C:	$a>0$ and $q<0$
D:	$a<0$ and $q>0$

ANSWERS: **Option**

Graph 1: D

Graph 2: A

Graph 3: B

(b) Consider $g(x)=\frac{9}{x}-3$.

(1) Determine the coordinates of the x-intercept(s) of g. (2)

$\frac{9}{x}-3=0$ ✓

$\frac{9}{x}=3$

$3x=9$

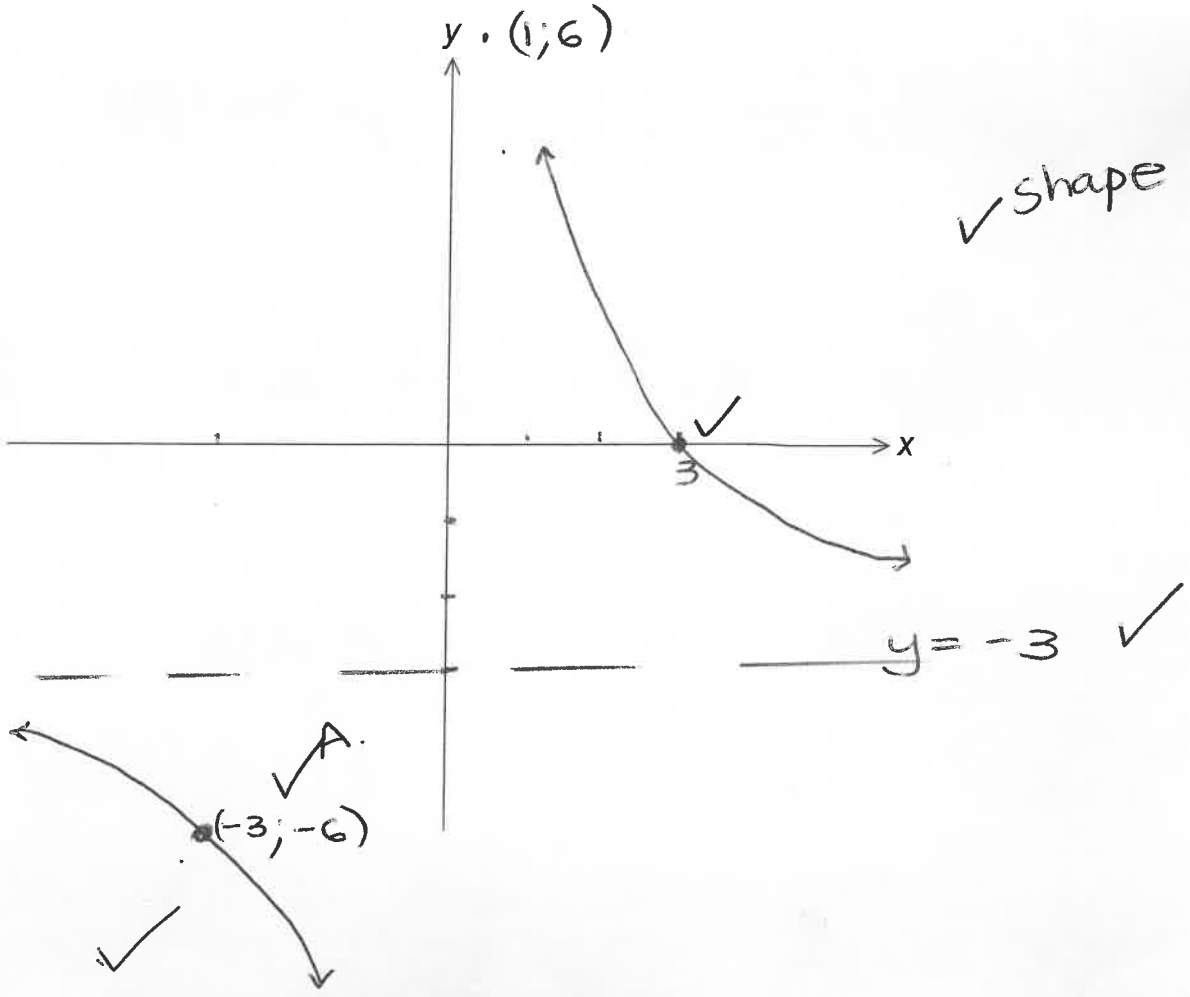
$x=3$ ✓

(2) Write down the equations of the asymptotes of g. (2)

$y=-3$ ✓

$x=0$ ✓

(4) Hence, or otherwise, make a neat sketch graph of g.
Label all pertinent information.
Indicate and label one miscellaneous point on your graph. (5)



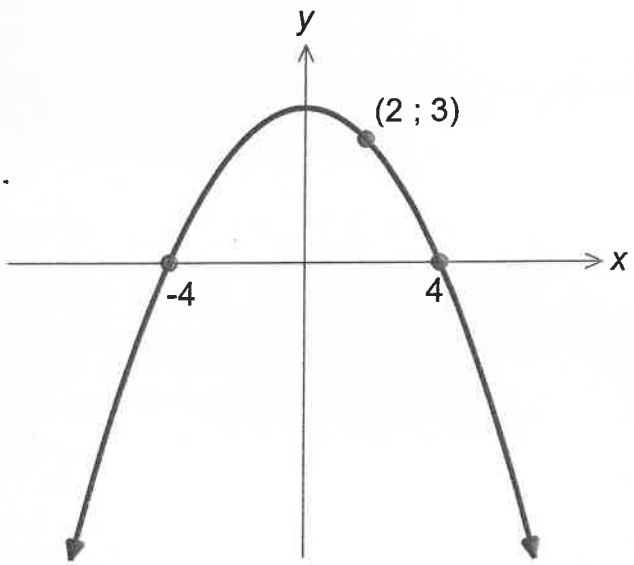
(5) One of the axes of symmetry of g has no points of intersection with g.
Write down the equation of this axis of symmetry. (2)

$y = -x + c$

$(0;-3)$ $\quad -3 = -0 + c$ (not necessary)

$\therefore y = -x - 3.$ ✓ ✓

(c) Determine the equation of the graph below. You need not simplify your answer. (4)



$y = a(x-4)(x+4)$

$y = a(x^2 - 16)$

Sub $(2;3)$ $\quad 3 = a(2^2 - 16)$ ✓

$3 = -12a$

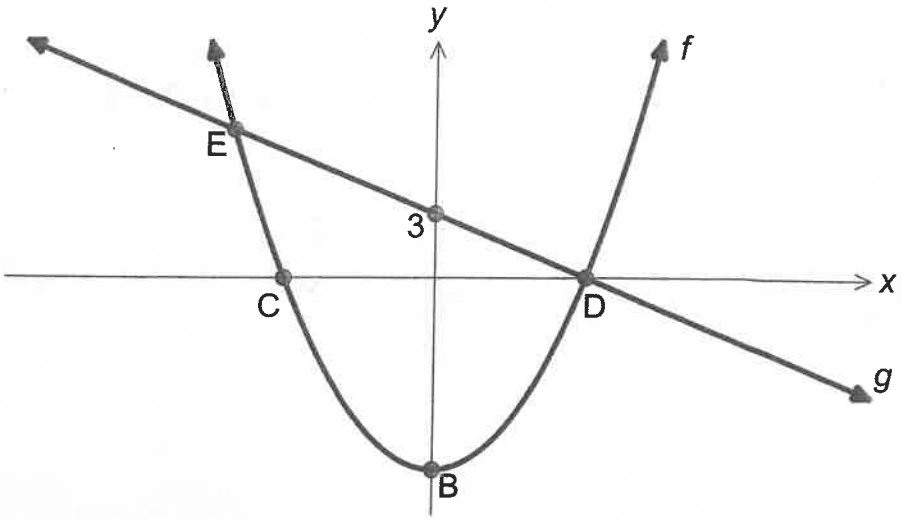
$a = -\frac{1}{4}$ ✓

$y = -\frac{1}{4}(x^2 - 16)$ ✓

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QUESTION 3

The graphs of $f(x) = x^2 - 9$ and $g(x) = mx + c$ are sketched below.



(a) Determine the co-ordinates of C and D. (3)

$x^2 - 9 = 0$ ✓

$x = \pm 3$

C (-3; 0) ✓ D (3; 0) ✓

(b) Write down the co-ordinates of B. (2)

(0; -9) ✓ ✓

(c) Determine the equation of g. (2)

$y = mx + 3$ ✓

Sub (3; 0) $0 = m(3) + 3$

$m = -1$ ✓

or

calc gradient: (3; 0) (0; 3)

$m = \frac{3-0}{0-3}$

$= -1$

(d) Determine the co-ordinates of E. (5)

$x^2 - 9 = -x + 3$ ✓

$x^2 + x - 12 = 0$ ✓

$(x+4)(x-3) = 0$

$x + 4 = 0$

$x = -4$ ✓

$y = -x + 3$ ✓

$= -(-4) + 3$

$= 7$

E (-4; 7) ✓

(e) Write down the domain of g. (1)

$x \in \mathbb{R}$. ✓

(f) Write down the range of f. (2)

$y \geq -9$ ✓ ✓

(g) For which values of x is f increasing? (1)

$x > 0$ ✓

(h) Use the graph to determine the values of x for which $f(x) \leq 0$. (2)

$-3 \leq x \leq 3$ ✓ ✓

(i) Determine the minimum value of $y = f(-x) + 5$ (2)

$y = -4$

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